



**Generation Interconnection
System Impact Study Report
for
Queue Project AF2-037
LOUISA-NORTH ANNA 230 KV
56.4 MW Capacity / 94 MW Energy**

February 2021

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Dominion.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Louisa County, Virginia. The installed facilities will have a total capability of 94 MW with 56.4 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is July 01, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-037
Project Name	LOUISA-NORTH ANNA 230 KV
State	Virginia
County	Louisa
Transmission Owner	Dominion
MFO	94
MWE	94
MWC	56.4
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF2-037 will interconnect with the Dominion on transmission system tapping the line between Louisa 230 kV substation and North Anna 230 kV substation. The IC is responsible for securing right-of-way, permits and constructing the proposed attachment line from the solar facility site to the proposed new substation. Attachment 1 shows a one-line diagram of the proposed interconnection facilities. The IC may not install any facilities on Dominion's right-of-way without first obtaining the necessary approval from Dominion Energy.

5 Cost Summary

The AF2-037 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$10,400,000
Allocation towards System Network Upgrade Costs*	\$17,666,221
Total Costs	\$28,066,221

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AF2-037. The project was evaluated as a 94 MW Capacity (94.0 MW energy) injection at a single line tap between Louisa 230 kV substation and North Anna 230 kV substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2023 AF2 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically, in Planning Studies, NERC Planning Event 3 and 6 Contingency Conditions (Loss of generator, transmission circuit, transformer, shunt device, or Single Pole of a DC line followed by the loss of a generator, transmission circuit, transformer, shunt device or single pole of a DC line) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

Note that the ITO findings were made from a conceptual review of this project and cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facilities	\$ 2,100,000
Three breaker 230kV ring-bus substation	\$ 6,500,000
Re-arrange line and tie-in new substation	\$ 1,800,000
Total Physical Interconnection Costs	\$ 10,400,000

It is estimated to take 24-30 months to complete this work upon execution of an Interconnection Construction Service Agreement (ICSA). These preliminary cost estimates are based on typical engineering costs. A more

detailed engineering cost estimates are normally done when the IC provides an exact site plan location for the generation substation during the Facility Study phase. See Attachment One.

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

7 Schedule

Based on the scope of work for the interconnection facilities, it is expected to take a minimum of **24-30 months** after the signing of an Interconnection Construction Service Agreement and construction kickoff call to complete the installation of the physical connection work. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined interconnection work, and that all system outages will be allowed when requested.

The schedule for any required Network Impact Reinforcements will be more clearly identified in future study phases. The estimated time to complete each of the required reinforcements is identified in the "System Reinforcements" section of the report.

8 Transmission Owner Analysis

8.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2023 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no further deficiencies were identified.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion's "Dominion Energy Electric Transmission Generator Interconnection Requirements" documented in Dominion's Facility Interconnection Requirements "Exhibit C" located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC's GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.

3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Irradiance (Watts/meter²) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Ambient air temperature (Fahrenheit) - (Accepted, not required)
- Wind speed (meters/second) - (Accepted, not required)
- Wind direction (decimal degrees from true north) - (Accepted, not required)

10.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

See Section 3.4.6 “Metering and telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

11 Summer Peak Analysis

The Queue Project AF2-037 was evaluated as a 94.0 MW (Capacity 56.4 MW) injection tapping the Louisa to North Anna 230 kV line in the Dominion area. Project AF2-037 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-037 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
96822699	314918	8NO ANNA	500.0	DVP	314911	8LADYSMIT H	500.0	DVP	1	DVP_P4 -2: SPOTS H1T594	breaker	3938.0	105.19	105.65	AC	46.29
96822700	314918	8NO ANNA	500.0	DVP	314911	8LADYSMIT H	500.0	DVP	1	DVP_P4 -2: H1T594	breaker	3938.0	101.27	101.67	AC	46.12
96822709	314918	8NO ANNA	500.0	DVP	314934	8SPOTSYL	500.0	DVP	1	DVP_P4 -2: 568T575	breaker	3938.0	103.15	103.45	AC	39.93
96822689	314934	8SPOTSYL	500.0	DVP	314916	8MORRSL	500.0	DVP	1	DVP_P4 -2: 568T575	breaker	3938.0	109.52	109.84	AC	39.47
97882815	957430	AF2-037 TAP	230.0	DVP	314232	6NO ANNA	230.0	DVP	1	DVP_P1 -2: LN 2088	single	749.18	111.53	118.55	AC	56.18

11.4 Steady-State Voltage Requirements

To be determined Facilities Study phase.

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE C T LOADIN G %	POST PROJE C T LOADIN G %	AC D C	MW IMPAC T
96822871	314918	8NO ANNA	500.0	DVP	314911	8LADYSMIT H	500.0	DVP	1	DVP_P1-2: LN 594	operatio n	3218.56	123.14	123.62	AC	45.83
96822881	314918	8NO ANNA	500.0	DVP	314934	8SPOTSYL	500.0	DVP	1	DVP_P1-2: LN 575	operatio n	3218.56	120.12	120.51	AC	39.58
96822854	314934	8SPOTSY L	500.0	DVP	314916	8MORRSVL	500.0	DVP	1	DVP_P1-2: LN 575	operatio n	3218.56	126.13	126.55	AC	39.07
97882814	957430	AF2-037 TAP	230.0	DVP	314232	6NO ANNA	230.0	DVP	1	DVP_P1-2: LN 2088	operatio n	749.18	125.68	137.51	AC	93.63

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-037	Upgrade Number												
97882815	2	AF2-037 TAP 230.0 kV - 6NO ANNA 230.0 kV Ckt 1	DVP ProjectId : n6773 (dom-083) Description : Replace Wavetraps at the Louisa and North Anna 230 kV substations.. Type : FAC Total Cost : \$300,000 Time Estimate : 16-18 Months Ratings : 872.0/872.0/1003.0 Queue Project AF2-037 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF2-037 could receive cost allocation. Note 2: Although Queue Project AF2-037 may not have cost responsibility for this upgrade, Queue Project AF2-037 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-037 comes into service prior to completion of the upgrade, Queue Project AF2-037 will need an interim study ProjectId : n6785 (dom-196) Description : Rebuild 15.97 miles of 230 kV Line 255 from Louisa to North Anna with 2-636 ACSR. Upgrade Relay and Wavetrap at North Anna. Type : FAC Total Cost : \$24,275,000 Time Estimate : 36-40 Months Ratings : 1047.0/1047.0/1204.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AF1-301</td><td>75.7</td><td>57.40%</td><td>\$13,934,012</td></tr><tr><td>AF2-037</td><td>56.18</td><td>42.60%</td><td>\$10,340,988</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-301	75.7	57.40%	\$13,934,012	AF2-037	56.18	42.60%	\$10,340,988	\$24,575,000	\$10,340,988	n6773, n6785
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AF1-301	75.7	57.40%	\$13,934,012															
AF2-037	56.18	42.60%	\$10,340,988															

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-037	Upgrade Number																																																																																
96822689	4	8SPOTSYL 500.0 kV - 8MORRSVL 500.0 kV Ckt 1	DVP ProjectId : n6160 (n6160) Description : Rebuild 18.75 miles of 500 kV Line 594 from Spotsylvania to Morrisville with 3-1351.5 113C ACSR. Type : FAC Total Cost : \$58,125,000 Time Estimate : 48-60 Months Ratings : 4453.0/4453.0/5121.0 <table><thead><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr></thead><tbody><tr><td>AE1-069</td><td>11.49</td><td>1.24%</td><td>\$723,234</td></tr><tr><td>AE1-173</td><td>178.09</td><td>19.29%</td><td>\$11,209,817</td></tr><tr><td>AE1-206</td><td>62.76</td><td>6.80%</td><td>\$3,950,408</td></tr><tr><td>AE2-031</td><td>38.03</td><td>4.12%</td><td>\$2,393,786</td></tr><tr><td>AE2-034</td><td>9.99</td><td>1.08%</td><td>\$628,817</td></tr><tr><td>AE2-094</td><td>45.55</td><td>4.93%</td><td>\$2,867,130</td></tr><tr><td>AE2-122</td><td>34.55</td><td>3.74%</td><td>\$2,174,738</td></tr><tr><td>AE2-123</td><td>35.51</td><td>3.85%</td><td>\$2,235,165</td></tr><tr><td>AE2-124</td><td>32.28</td><td>3.50%</td><td>\$2,031,854</td></tr><tr><td>AE2-250</td><td>8.73</td><td>0.95%</td><td>\$549,507</td></tr><tr><td>AE2-270</td><td>32.6</td><td>3.53%</td><td>\$2,051,996</td></tr><tr><td>AE2-313</td><td>41.13</td><td>4.45%</td><td>\$2,588,914</td></tr><tr><td>AF1-123</td><td>55.44</td><td>6.00%</td><td>\$3,489,653</td></tr><tr><td>AF1-124</td><td>55.44</td><td>6.00%</td><td>\$3,489,653</td></tr><tr><td>AF1-125</td><td>55.44</td><td>6.00%</td><td>\$3,489,653</td></tr><tr><td>AF1-128</td><td>54.08</td><td>5.86%</td><td>\$3,404,048</td></tr><tr><td>AF1-236</td><td>87.71</td><td>9.50%</td><td>\$5,520,877</td></tr><tr><td>AF1-301</td><td>45.14</td><td>4.89%</td><td>\$2,841,323</td></tr><tr><td>AF2-037</td><td>39.47</td><td>4.27%</td><td>\$2,484,426</td></tr></tbody></table>	Queue	MW	Cost %	Cost \$	AE1-069	11.49	1.24%	\$723,234	AE1-173	178.09	19.29%	\$11,209,817	AE1-206	62.76	6.80%	\$3,950,408	AE2-031	38.03	4.12%	\$2,393,786	AE2-034	9.99	1.08%	\$628,817	AE2-094	45.55	4.93%	\$2,867,130	AE2-122	34.55	3.74%	\$2,174,738	AE2-123	35.51	3.85%	\$2,235,165	AE2-124	32.28	3.50%	\$2,031,854	AE2-250	8.73	0.95%	\$549,507	AE2-270	32.6	3.53%	\$2,051,996	AE2-313	41.13	4.45%	\$2,588,914	AF1-123	55.44	6.00%	\$3,489,653	AF1-124	55.44	6.00%	\$3,489,653	AF1-125	55.44	6.00%	\$3,489,653	AF1-128	54.08	5.86%	\$3,404,048	AF1-236	87.71	9.50%	\$5,520,877	AF1-301	45.14	4.89%	\$2,841,323	AF2-037	39.47	4.27%	\$2,484,426	\$58,125,000	\$2,484,426	n6160
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AF1-301	45.14	4.89%	\$2,841,323																																																																																			
AF2-037	39.47	4.27%	\$2,484,426																																																																																			
96822699,96822700	3	8NO ANNA 500.0 kV - 8LADYSMITH 500.0 kV Ckt 1	DVP ProjectId : n6574 (dom-044) Description : Rebuild 14.53 miles of 500 kV Line 575 from North Anna to Ladysmith with 3-1351.5 113C ACSR. Type : FAC Total Cost : \$45,043,000 Time Estimate : 48-60 Months Ratings : 4453.0/4453.0/5121.0 <table><thead><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr></thead><tbody><tr><td>AE2-031</td><td>34.63</td><td>4.59%</td><td>\$2,069,191</td></tr><tr><td>AE2-094</td><td>41.5</td><td>5.51%</td><td>\$2,479,683</td></tr><tr><td>AE2-122</td><td>115.02</td><td>15.26%</td><td>\$6,872,607</td></tr><tr><td>AE2-123</td><td>115.02</td><td>15.26%</td><td>\$6,872,607</td></tr><tr><td>AE2-124</td><td>115.08</td><td>15.27%</td><td>\$6,876,192</td></tr><tr><td>AE2-313</td><td>62.52</td><td>8.29%</td><td>\$3,735,658</td></tr><tr><td>AF1-123</td><td>34.98</td><td>4.64%</td><td>\$2,090,104</td></tr><tr><td>AF1-124</td><td>37.37</td><td>4.96%</td><td>\$2,232,910</td></tr><tr><td>AF1-125</td><td>37.37</td><td>4.96%</td><td>\$2,232,910</td></tr><tr><td>AF1-236</td><td>61.48</td><td>8.16%</td><td>\$3,673,516</td></tr><tr><td>AF1-301</td><td>52.58</td><td>6.97%</td><td>\$3,141,729</td></tr><tr><td>AF2-037</td><td>46.29</td><td>6.14%</td><td>\$2,765,893</td></tr></tbody></table>	Queue	MW	Cost %	Cost \$	AE2-031	34.63	4.59%	\$2,069,191	AE2-094	41.5	5.51%	\$2,479,683	AE2-122	115.02	15.26%	\$6,872,607	AE2-123	115.02	15.26%	\$6,872,607	AE2-124	115.08	15.27%	\$6,876,192	AE2-313	62.52	8.29%	\$3,735,658	AF1-123	34.98	4.64%	\$2,090,104	AF1-124	37.37	4.96%	\$2,232,910	AF1-125	37.37	4.96%	\$2,232,910	AF1-236	61.48	8.16%	\$3,673,516	AF1-301	52.58	6.97%	\$3,141,729	AF2-037	46.29	6.14%	\$2,765,893	\$45,043,000	\$2,765,893	n6574																												
Queue	MW	Cost %	Cost \$																																																																																			
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ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-037	Upgrade Number																																																																								
96822709	1	8NO ANNA 500.0 kV - 8SPOTSYL 500.0 kV Ckt 1	DVP ProjectId : n6132 (n6132) Description : Rebuild 14.02 miles of 500 kV Line 573 from North Anna to Spotsylvania with 3-1351.5 113C ACSR. Type : FAC Total Cost : \$43,462,000 Time Estimate : 48-60 Months Ratings : 4453.0/4453.0/5121.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE1-075</td><td>2.36</td><td>0.28%</td><td>\$122,635</td></tr><tr><td>AE1-173</td><td>108.39</td><td>12.96%</td><td>\$5,632,356</td></tr><tr><td>AE1-206</td><td>63.74</td><td>7.62%</td><td>\$3,312,172</td></tr><tr><td>AE2-031</td><td>38.46</td><td>4.60%</td><td>\$1,998,528</td></tr><tr><td>AE2-094</td><td>46.07</td><td>5.51%</td><td>\$2,393,972</td></tr><tr><td>AE2-122</td><td>34.93</td><td>4.18%</td><td>\$1,815,095</td></tr><tr><td>AE2-123</td><td>35.9</td><td>4.29%</td><td>\$1,865,500</td></tr><tr><td>AE2-124</td><td>32.64</td><td>3.90%</td><td>\$1,696,098</td></tr><tr><td>AE2-270</td><td>32.95</td><td>3.94%</td><td>\$1,712,207</td></tr><tr><td>AE2-313</td><td>41.59</td><td>4.97%</td><td>\$2,161,174</td></tr><tr><td>AF1-123</td><td>56.62</td><td>6.77%</td><td>\$2,942,190</td></tr><tr><td>AF1-124</td><td>56.62</td><td>6.77%</td><td>\$2,942,190</td></tr><tr><td>AF1-125</td><td>56.62</td><td>6.77%</td><td>\$2,942,190</td></tr><tr><td>AF1-128</td><td>55.21</td><td>6.60%</td><td>\$2,868,921</td></tr><tr><td>AF1-236</td><td>89.56</td><td>10.71%</td><td>\$4,653,878</td></tr><tr><td>AF1-301</td><td>44.8</td><td>5.36%</td><td>\$2,327,978</td></tr><tr><td>AF2-037</td><td>39.93</td><td>4.77%</td><td>\$2,074,914</td></tr></table>	Queue	MW	Cost %	Cost \$	AE1-075	2.36	0.28%	\$122,635	AE1-173	108.39	12.96%	\$5,632,356	AE1-206	63.74	7.62%	\$3,312,172	AE2-031	38.46	4.60%	\$1,998,528	AE2-094	46.07	5.51%	\$2,393,972	AE2-122	34.93	4.18%	\$1,815,095	AE2-123	35.9	4.29%	\$1,865,500	AE2-124	32.64	3.90%	\$1,696,098	AE2-270	32.95	3.94%	\$1,712,207	AE2-313	41.59	4.97%	\$2,161,174	AF1-123	56.62	6.77%	\$2,942,190	AF1-124	56.62	6.77%	\$2,942,190	AF1-125	56.62	6.77%	\$2,942,190	AF1-128	55.21	6.60%	\$2,868,921	AF1-236	89.56	10.71%	\$4,653,878	AF1-301	44.8	5.36%	\$2,327,978	AF2-037	39.93	4.77%	\$2,074,914	\$43,462,000	\$2,074,914	n6132
Queue	MW	Cost %	Cost \$																																																																											
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			TOTAL COST	\$171,205,000	\$17,666,221																																																																									

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96822709	314918	8NO ANNA	DVP	314934	8SPOTSYL	DVP	1	DVP_P4-2: 568T575	breaker	3938.0	103.15	103.45	AC	39.93

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
314333	6POWHATN	0.4601	50/50	0.4601
314435	6SAPONY	0.4243	50/50	0.4243
314947	8GREENSVILLE	58.2620	50/50	58.2620
315102	1BRUNSWICKG1	9.5681	50/50	9.5681
315103	1BRUNSWICKG2	9.5681	50/50	9.5681
315104	1BRUNSWICKG3	9.5681	50/50	9.5681
315105	1BRUNSWICKS1	19.8777	50/50	19.8777
315153	1CLOVER1	13.6464	50/50	13.6464
315154	1CLOVER2	13.5104	50/50	13.5104
315177	1S ANNAG1	1.6978	50/50	1.6978
315178	1S ANNAS1	0.8725	50/50	0.8725
315179	1S ANNAG2	1.6978	50/50	1.6978
315180	1S ANNAS2	0.8725	50/50	0.8725
315225	1N ANNA1	96.4965	50/50	96.4965
315226	1N ANNA2	96.5372	50/50	96.5372
916192	Z1-068 E	1.6035	Adder	1.89
923852	AB2-025 E	0.9644	Adder	1.13
923862	AB2-026 E	1.2300	50/50	1.2300
925021	AB2-158 C	3.5662	50/50	3.5662
925022	AB2-158 E	9.4632	50/50	9.4632
926751	AC1-161 C O1	26.0105	Adder	30.6
926752	AC1-161 E O1	11.1032	Adder	13.06
933291	AC2-141 C	26.0105	Adder	30.6
933292	AC2-141 E	11.1032	Adder	13.06
933501	AC2-165 C	10.7969	50/50	10.7969
933502	AC2-165 E	8.1261	50/50	8.1261
933731	AC2-196 C	1.5298	Adder	1.8
933732	AC2-196 E	1.0189	Adder	1.2
934061	AD1-033 C	6.4235	Adder	7.56
934062	AD1-033 E	4.2823	Adder	5.04
938491	AE1-068 C O1	69.1452	50/50	69.1452
938492	AE1-068 E O1	38.1898	50/50	38.1898
938501	AE1-069 C O1	54.1347	50/50	54.1347
938502	AE1-069 E O1	30.9493	50/50	30.9493
938551	AE1-074 C	2.5936	50/50	2.5936
938552	AE1-074 E	1.3066	50/50	1.3066
938561	AE1-075 C	2.4005	50/50	2.4005
938562	AE1-075 E	1.1705	50/50	1.1705
939231	AE1-154 C	3.4791	50/50	3.4791
939232	AE1-154 E	2.4354	50/50	2.4354
939411	AE1-173 C	100.1904	50/50	100.1904

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939412	AE1-173 E	66.7936	50/50	66.7936
940251	AE2-007 O1	76.7072	Adder	90.24
940471	AE2-031 C	37.7510	50/50	37.7510
940472	AE2-031 E	25.1674	50/50	25.1674
940481	AE2-033 C	13.7517	Adder	16.18
940482	AE2-033 E	9.2708	Adder	10.91
940541	AE2-040 O1	2.4420	Adder	2.87
940641	AE2-051 C O1	19.6434	50/50	19.6434
940642	AE2-051 E O1	13.0956	50/50	13.0956
941031	AE2-094 C	45.2442	50/50	45.2442
941032	AE2-094 E	20.2638	50/50	20.2638
941281	AE2-122 C O1	24.2685	Adder	28.55
941282	AE2-122 E O1	97.9292	Adder	115.21
941291	AE2-123 C O1	24.9405	Adder	29.34
941292	AE2-123 E O1	97.2572	Adder	114.42
941301	AE2-124 C O1	22.6852	Adder	26.69
941302	AE2-124 E O1	99.5397	Adder	117.11
942931	AE2-313 C	40.8753	50/50	40.8753
942932	AE2-313 E	27.2502	50/50	27.2502
944011	AF1-069 C	14.7830	50/50	14.7830
944012	AF1-069 E	5.7429	50/50	5.7429
944111	AF1-079 C	6.6103	50/50	6.6103
944112	AF1-079 E	8.9761	50/50	8.9761
944581	AF1-123 C O1	41.2572	Adder	48.54
944582	AF1-123 E O1	94.4674	Adder	111.14
944591	AF1-124 C O1	41.2572	Adder	48.54
944592	AF1-124 E O1	94.4674	Adder	111.14
944601	AF1-125 C O1	41.2572	Adder	48.54
944602	AF1-125 E O1	94.4674	Adder	111.14
945811	AF1-246 C O1	8.1241	50/50	8.1241
945812	AF1-246 E O1	11.2189	50/50	11.2189
946011	AF1-266	11.5112	Adder	13.54
946371	AF1-301 C	26.8721	50/50	26.8721
946372	AF1-301 E	18.0326	50/50	18.0326
957431	AF2-037 C	23.9576	50/50	23.9576
957432	AF2-037 E	15.9717	50/50	15.9717
957481	AF2-042 C	58.0290	50/50	58.0290
957482	AF2-042 E	38.6860	50/50	38.6860
957691	AF2-063 C	38.2302	50/50	38.2302
957692	AF2-063 E	25.4868	50/50	25.4868
957911	AF2-085	4.6456	50/50	4.6456
958501	AF2-144 C	2.0236	50/50	2.0236
958502	AF2-144 E	1.3491	50/50	1.3491
959531	AF2-244 C	1.8649	50/50	1.8649
959532	AF2-244 E	1.1110	50/50	1.1110
959751	AF2-266	9.1970	50/50	9.1970
WEC	WEC	0.7119	Confirmed LTF	0.7119
LGEE	LGEE	1.4086	Confirmed LTF	1.4086
CPL	CPL	8.1675	Confirmed LTF	8.1675
CBM-W2	CBM-W2	31.2121	Confirmed LTF	31.2121
NY	NY	2.6627	Confirmed LTF	2.6627
TVA	TVA	6.7886	Confirmed LTF	6.7886

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
O-066	O-066	36.1334	Confirmed LTF	36.1334
CBM-S2	CBM-S2	56.5804	Confirmed LTF	56.5804
CBM-S1	CBM-S1	37.5476	Confirmed LTF	37.5476
G-007	G-007	5.6326	Confirmed LTF	5.6326
MADISON	MADISON	2.0100	Confirmed LTF	2.0100
MEC	MEC	4.5048	Confirmed LTF	4.5048
AA2-074	AA2-074	5.5661	LTF	5.5661
CBM-W1	CBM-W1	25.5704	Confirmed LTF	25.5704

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
97882815	957430	AF2-037 TAP	DVP	314232	6NO ANNA	DVP	1	DVP_P1-2: LN 2088	single	749.18	111.53	118.55	AC	56.18

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
315172	1LOISA A	6.5476	80/20	6.5476
315173	1LOISA B	6.5816	80/20	6.5816
315174	1LOISA C	6.5816	80/20	6.5816
315175	1LOISA D	6.5816	80/20	6.5816
315176	1LOISA E	13.4178	80/20	13.4178
315177	1S ANNAG1	6.1145	80/20	6.1145
315178	1S ANNAS1	3.1422	80/20	3.1422
315179	1S ANNAG2	6.1145	80/20	6.1145
315180	1S ANNAS2	3.1422	80/20	3.1422
925021	AB2-158 C	5.1803	80/20	5.1803
939231	AE1-154 C	9.9609	80/20	9.9609
944111	AF1-079 C	18.9257	80/20	18.9257
946371	AF1-301 C	75.7028	80/20	75.7028
957431	AF2-037 C	56.1795	80/20	56.1795
957691	AF2-063 C	89.6481	80/20	89.6481
NEWTON	NEWTON	0.4201	Confirmed LTF	0.4201
FARMERCITY	FARMERCITY	0.0219	Confirmed LTF	0.0219
GIBSON	GIBSON	0.2135	Confirmed LTF	0.2135
NY	NY	0.2190	Confirmed LTF	0.2190
PRAIRIE	PRAIRIE	1.0100	Confirmed LTF	1.0100
COFFEEN	COFFEEN	0.0780	Confirmed LTF	0.0780
CHEOAH	CHEOAH	0.1957	Confirmed LTF	0.1957
EDWARDS	EDWARDS	0.1368	Confirmed LTF	0.1368
TILTON	TILTON	0.2463	Confirmed LTF	0.2463
CALDERWOOD	CALDERWOOD	0.1943	Confirmed LTF	0.1943
BLUEG	BLUEG	0.6788	Confirmed LTF	0.6788
TRIMBLE	TRIMBLE	0.2176	Confirmed LTF	0.2176
CATAWBA	CATAWBA	0.1368	Confirmed LTF	0.1368

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96822699	314918	8NO ANNA	DVP	314911	8LADYSMITH	DVP	1	DVP_P4-2: SPOTS H1T594	breaker	3938.0	105.19	105.65	AC	46.29

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
315172	1LOISA A	2.9434	50/50	2.9434
315173	1LOISA B	2.9587	50/50	2.9587
315174	1LOISA C	2.9587	50/50	2.9587
315175	1LOISA D	2.9587	50/50	2.9587
315176	1LOISA E	6.0319	50/50	6.0319
315177	1S ANNAG1	2.0970	50/50	2.0970
315178	1S ANNAS1	1.0776	50/50	1.0776
315179	1S ANNAG2	2.0970	50/50	2.0970
315180	1S ANNAS2	1.0776	50/50	1.0776
315225	1N ANNA1	109.9900	50/50	109.9900
315226	1N ANNA2	110.0364	50/50	110.0364
916192	Z1-068 E	1.2220	Adder	1.44
919152	AA1-139 E	4.1946	Adder	4.93
923852	AB2-025 E	0.7714	Adder	0.91
923862	AB2-026 E	0.9274	Adder	1.09
924032	AB2-045 E	0.7322	Adder	0.86
924512	AB2-100 E	3.9166	Adder	4.61
925021	AB2-158 C	4.1851	50/50	4.1851
925022	AB2-158 E	11.1055	50/50	11.1055
925522	AC1-027 E	0.7444	Adder	0.88
926731	AC1-158 C	40.2933	50/50	40.2933
926732	AC1-158 E	105.2296	50/50	105.2296
926751	AC1-161 C O1	19.8114	Adder	23.31
926752	AC1-161 E O1	8.4569	Adder	9.95
932511	AC2-071 C	1.5458	Adder	1.82
932512	AC2-071 E	0.7787	Adder	0.92
933291	AC2-141 C	19.8114	Adder	23.31
933292	AC2-141 E	8.4569	Adder	9.95
933501	AC2-165 C	8.1498	Adder	9.59
933502	AC2-165 E	6.1338	Adder	7.22
933731	AC2-196 C	1.1691	Adder	1.38
933732	AC2-196 E	0.7786	Adder	0.92
934061	AD1-033 C	4.9073	Adder	5.77
934062	AD1-033 E	3.2715	Adder	3.85
934611	AD1-087 C O1	6.1677	Adder	7.26
934612	AD1-087 E O1	2.8987	Adder	3.41
934621	AD1-088 C	8.9949	Adder	10.58
934622	AD1-088 E	4.2329	Adder	4.98
935171	AD1-152 C O1	6.1294	Adder	7.21
935172	AD1-152 E O1	4.0863	Adder	4.81
935221	AD1-157 C	1.0467	Adder	1.23

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
935222	AD1-157 E	0.6978	Adder	0.82
937221	AD2-160 C O1	3.8254	Adder	4.5
937222	AD2-160 E O1	2.0060	Adder	2.36
937481	AD2-202 C O1	1.6345	Adder	1.92
937482	AD2-202 E O1	0.9194	Adder	1.08
938371	AE1-056 C	4.5127	Adder	5.31
938372	AE1-056 E	2.4657	Adder	2.9
938491	AE1-068 C O1	52.5010	Adder	61.77
938492	AE1-068 E O1	28.9970	Adder	34.11
938501	AE1-069 C O1	41.0758	Adder	48.32
938502	AE1-069 E O1	23.4834	Adder	27.63
938531	AE1-072 C O1	11.4769	Adder	13.5
938532	AE1-072 E O1	5.9829	Adder	7.04
938551	AE1-074 C	1.7260	Adder	2.03
938552	AE1-074 E	0.8695	Adder	1.02
938561	AE1-075 C	1.8099	Adder	2.13
938562	AE1-075 E	0.8825	Adder	1.04
939231	AE1-154 C	4.0829	50/50	4.0829
939232	AE1-154 E	2.8580	50/50	2.8580
939411	AE1-173 C	72.5546	Adder	85.36
939412	AE1-173 E	48.3698	Adder	56.91
940251	AE2-007 O1	76.1603	Adder	89.6
940471	AE2-031 C	28.6882	Adder	33.75
940472	AE2-031 E	19.1254	Adder	22.5
940481	AE2-033 C	11.0063	Adder	12.95
940482	AE2-033 E	7.4200	Adder	8.73
940541	AE2-040 O1	1.9533	Adder	2.3
940641	AE2-051 C O1	14.6253	Adder	17.21
940642	AE2-051 E O1	9.7502	Adder	11.47
941031	AE2-094 C	34.3962	Adder	40.47
941032	AE2-094 E	15.4053	Adder	18.12
941281	AE2-122 C O1	18.4999	Adder	21.76
941282	AE2-122 E O1	74.6514	Adder	87.83
941291	AE2-123 C O1	19.0121	Adder	22.37
941292	AE2-123 E O1	74.1391	Adder	87.22
941301	AE2-124 C O1	17.2979	Adder	20.35
941302	AE2-124 E O1	75.9010	Adder	89.3
941791	AE2-182 C	1.4614	Adder	1.72
941792	AE2-182 E	0.6803	Adder	0.8
942401	AE2-253 C	4.0237	Adder	4.73
942402	AE2-253 E	1.8077	Adder	2.13
942461	AE2-259 C O1	7.0773	Adder	8.33
942462	AE2-259 E O1	4.7182	Adder	5.55
942471	AE2-260 C O1	9.8701	Adder	11.61
942472	AE2-260 E O1	13.9996	Adder	16.47
942931	AE2-313 C	31.0624	Adder	36.54
942932	AE2-313 E	20.7082	Adder	24.36
944011	AF1-069 C	11.2385	Adder	13.22
944012	AF1-069 E	4.3659	Adder	5.14
944111	AF1-079 C	7.7575	50/50	7.7575
944112	AF1-079 E	10.5339	50/50	10.5339
944581	AF1-123 C O1	31.6370	Adder	37.22

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944582	AF1-123 E O1	72.4398	Adder	85.22
944591	AF1-124 C O1	31.6370	Adder	37.22
944592	AF1-124 E O1	72.4398	Adder	85.22
944601	AF1-125 C O1	31.6370	Adder	37.22
944602	AF1-125 E O1	72.4398	Adder	85.22
945811	AF1-246 C O1	6.1258	Adder	7.21
945812	AF1-246 E O1	8.4594	Adder	9.95
946011	AF1-266	9.2132	Adder	10.84
946281	AF1-292 C	1.0382	Adder	1.22
946282	AF1-292 E	0.6999	Adder	0.82
946371	AF1-301 C	31.5020	50/50	31.5020
946372	AF1-301 E	21.1395	50/50	21.1395
957431	AF2-037 C	27.7725	50/50	27.7725
957432	AF2-037 E	18.5150	50/50	18.5150
957481	AF2-042 C	43.7555	Adder	51.48
957482	AF2-042 E	29.1703	Adder	34.32
957691	AF2-063 C	44.3178	50/50	44.3178
957692	AF2-063 E	29.5452	50/50	29.5452
957871	AF2-081 C	6.5250	Adder	7.68
957872	AF2-081 E	2.7964	Adder	3.29
957911	AF2-085	3.4896	Adder	4.11
958501	AF2-144 C	1.5257	Adder	1.79
958502	AF2-144 E	1.0171	Adder	1.2
959531	AF2-244 C	1.4060	Adder	1.65
959532	AF2-244 E	0.8376	Adder	0.99
959751	AF2-266	6.9096	Adder	8.13
960081	AF2-299 C	1.7498	Adder	2.06
960082	AF2-299 E	1.1665	Adder	1.37
WEC	WEC	0.5311	Confirmed LTF	0.5311
LGEE	LGEE	1.0526	Confirmed LTF	1.0526
CPL	CPL	6.8981	Confirmed LTF	6.8981
CBM-W2	CBM-W2	24.5782	Confirmed LTF	24.5782
NY	NY	2.3514	Confirmed LTF	2.3514
TVA	TVA	5.4236	Confirmed LTF	5.4236
O-066	O-066	31.8998	Confirmed LTF	31.8998
CBM-S2	CBM-S2	47.2804	Confirmed LTF	47.2804
CBM-S1	CBM-S1	29.7944	Confirmed LTF	29.7944
G-007	G-007	4.9733	Confirmed LTF	4.9733
MADISON	MADISON	1.7217	Confirmed LTF	1.7217
MEC	MEC	3.4529	Confirmed LTF	3.4529
AA2-074	AA2-074	4.6993	LTF	4.6993
CBM-W1	CBM-W1	18.9026	Confirmed LTF	18.9026

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96822689	314934	8SPOTSYL	DVP	314916	8MORRSVL	DVP	1	DVP_P4-2: 568T575	breaker	3938.0	109.52	109.84	AC	39.47

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
314947	8GREENSVILLE	56.9637	50/50	56.9637
315102	1BRUNSWICKG1	9.3559	50/50	9.3559
315103	1BRUNSWICKG2	9.3559	50/50	9.3559
315104	1BRUNSWICKG3	9.3559	50/50	9.3559
315105	1BRUNSWICKS1	19.4368	50/50	19.4368
315175	1LOISA D	2.6474	50/50	2.6474
315176	1LOISA E	5.3974	50/50	5.3974
315177	1S ANNAG1	1.8664	50/50	1.8664
315178	1S ANNAS1	0.9591	50/50	0.9591
315179	1S ANNAG2	1.8664	50/50	1.8664
315180	1S ANNAS2	0.9591	50/50	0.9591
315225	1N ANNA1	92.6457	50/50	92.6457
315226	1N ANNA2	92.6848	50/50	92.6848
916192	Z1-068 E	1.5812	Adder	1.86
923852	AB2-025 E	0.9489	Adder	1.12
923862	AB2-026 E	1.0337	Adder	1.22
925021	AB2-158 C	3.5996	50/50	3.5996
925022	AB2-158 E	9.5518	50/50	9.5518
926001	AC1-076 C	6.8156	50/50	6.8156
926002	AC1-076 E	11.0825	50/50	11.0825
926481	AC1-120 C O1	6.1426	Adder	7.23
926482	AC1-120 E O1	3.1644	Adder	3.72
926501	AC1-121 C O1	2.1096	Adder	2.48
926502	AC1-121 E O1	0.9927	Adder	1.17
926731	AC1-158 C	40.2156	50/50	40.2156
926732	AC1-158 E	105.0268	50/50	105.0268
926751	AC1-161 C O1	25.6474	Adder	30.17
926752	AC1-161 E O1	10.9482	Adder	12.88
933291	AC2-141 C	25.6474	Adder	30.17
933292	AC2-141 E	10.9482	Adder	12.88
933501	AC2-165 C	9.0984	Adder	10.7
933502	AC2-165 E	6.8478	Adder	8.06
933731	AC2-196 C	1.5083	Adder	1.77
933732	AC2-196 E	1.0045	Adder	1.18
934061	AD1-033 C	6.3332	Adder	7.45
934062	AD1-033 E	4.2221	Adder	4.97
938491	AE1-068 C O1	67.6023	50/50	67.6023
938492	AE1-068 E O1	37.3376	50/50	37.3376
938501	AE1-069 C O1	52.9284	50/50	52.9284
938502	AE1-069 E O1	30.2596	50/50	30.2596
938551	AE1-074 C	2.1743	Adder	2.56

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938552	AE1-074 E	1.0953	Adder	1.29
938561	AE1-075 C	2.0173	Adder	2.37
938562	AE1-075 E	0.9836	Adder	1.16
939231	AE1-154 C	3.5117	50/50	3.5117
939232	AE1-154 E	2.4582	50/50	2.4582
939411	AE1-173 C	98.2080	50/50	98.2080
939412	AE1-173 E	65.4720	50/50	65.4720
940251	AE2-007 O1	57.4565	Adder	67.6
940471	AE2-031 C	36.9106	50/50	36.9106
940472	AE2-031 E	24.6071	50/50	24.6071
940481	AE2-033 C	13.5300	Adder	15.92
940482	AE2-033 E	9.1214	Adder	10.73
940541	AE2-040 O1	2.4028	Adder	2.83
940641	AE2-051 C O1	19.2240	50/50	19.2240
940642	AE2-051 E O1	12.8160	50/50	12.8160
941031	AE2-094 C	44.2289	50/50	44.2289
941032	AE2-094 E	19.8091	50/50	19.8091
941281	AE2-122 C O1	23.9308	Adder	28.15
941282	AE2-122 E O1	96.5666	Adder	113.61
941291	AE2-123 C O1	24.5935	Adder	28.93
941292	AE2-123 E O1	95.9040	Adder	112.83
941301	AE2-124 C O1	22.3683	Adder	26.32
941302	AE2-124 E O1	98.1495	Adder	115.47
941381	AE2-134 (Suspended)	4.6965	50/50	4.6965
942931	AE2-313 C	39.9653	50/50	39.9653
942932	AE2-313 E	26.6435	50/50	26.6435
944011	AF1-069 C	14.4512	50/50	14.4512
944012	AF1-069 E	5.6140	50/50	5.6140
944111	AF1-079 C	6.6722	50/50	6.6722
944112	AF1-079 E	9.0602	50/50	9.0602
944581	AF1-123 C O1	40.6706	Adder	47.85
944582	AF1-123 E O1	93.1242	Adder	109.56
944591	AF1-124 C O1	40.6706	Adder	47.85
944592	AF1-124 E O1	93.1242	Adder	109.56
944601	AF1-125 C O1	40.6706	Adder	47.85
944602	AF1-125 E O1	93.1242	Adder	109.56
945811	AF1-246 C O1	6.7669	Adder	7.96
945812	AF1-246 E O1	9.3448	Adder	10.99
946011	AF1-266	11.3257	Adder	13.32
946371	AF1-301 C	27.0735	50/50	27.0735
946372	AF1-301 E	18.1677	50/50	18.1677
957431	AF2-037 C	23.6801	50/50	23.6801
957432	AF2-037 E	15.7867	50/50	15.7867
957481	AF2-042 C	48.3352	Adder	56.86
957482	AF2-042 E	32.2235	Adder	37.91
957691	AF2-063 C	37.7874	50/50	37.7874
957692	AF2-063 E	25.1916	50/50	25.1916
957911	AF2-085	4.5538	50/50	4.5538
958501	AF2-144 C	1.7005	Adder	2.0
958502	AF2-144 E	1.1337	Adder	1.33
959531	AF2-244 C	1.5672	Adder	1.84
959532	AF2-244 E	0.9336	Adder	1.1

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
959751	AF2-266	7.6696	Adder	9.02
WEC	WEC	0.6631	Confirmed LTF	0.6631
LGEE	LGEE	1.3224	Confirmed LTF	1.3224
CPL	CPL	7.9969	Confirmed LTF	7.9969
CBM-W2	CBM-W2	29.8853	Confirmed LTF	29.8853
NY	NY	2.7733	Confirmed LTF	2.7733
TVA	TVA	6.5576	Confirmed LTF	6.5576
O-066	O-066	37.4976	Confirmed LTF	37.4976
CBM-S2	CBM-S2	55.3088	Confirmed LTF	55.3088
CBM-S1	CBM-S1	36.1589	Confirmed LTF	36.1589
G-007	G-007	5.8438	Confirmed LTF	5.8438
MADISON	MADISON	1.9938	Confirmed LTF	1.9938
MEC	MEC	4.2553	Confirmed LTF	4.2553
AA2-074	AA2-074	5.4491	LTF	5.4491
CBM-W1	CBM-W1	23.6314	Confirmed LTF	23.6314

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-139	Hickory-Shawboro 230kV	In Service
AA2-074	CPL-PJM	Confirmed
AB2-025	Sapony 34.5kV	In Service
AB2-026	Powhatan 34.5kV	In Service
AB2-045	Buckingham 34.5kV	In Service
AB2-100	Clubhouse-Lakeview 230kV	Under Construction
AB2-158	Louisa-South Anna 230kV	Under Construction
AC1-027	Pendleton 34.5kV	In Service
AC1-076	Locust Grove-Paytes 115kV	Engineering and Procurement
AC1-120	Mitchell-Mountain Run 115kV	Engineering and Procurement
AC1-121	Mitchell-Mountain Run 115kV	Engineering and Procurement
AC1-158	Spotsylvania 500kV	Partially in Service - Under Construction
AC1-161	Septa 500kV	Engineering and Procurement
AC2-071	Buckingham 35kV	Engineering and Procurement
AC2-141	Septa 500kV	Active
AC2-165	Bremo-Powhatan 230kV	Active
AC2-196	Fentress 34.5kV	Engineering and Procurement
AD1-033	Fentress-Landstown 230 kV	Engineering and Procurement
AD1-087	Clover-Sedge Hill 230 kV	Active
AD1-088	Briery-Clover 230 kV	Active
AD1-152	Clover-Sedge Hill 230 kV	Active
AD1-157	South Creek 34.5 kV	Engineering and Procurement
AD2-160	Hickory-Moyock 230kV	Active
AD2-202	Clover-Sedge Hill 230kV	Active
AE1-056	Red House-South Creek 115 kV	Active
AE1-068	Carson-Rogers Rd 500 kV	Active
AE1-069	Carson-Rogers Road 500 kV	Active
AE1-072	Shawboro-Sligo 230 kV	Active
AE1-074	Winterpock 34.5 kV	Engineering and Procurement
AE1-075	Powhatan 34.5 kV	Engineering and Procurement
AE1-154	Louisa-South Anna 230 kV	Engineering and Procurement
AE1-173	Carson-Suffolk 500 kV	Active
AE2-007	Chesapeake 230 kV	Active
AE2-031	Carson-Rawlings 500 kV	Active
AE2-033	Clubhouse-Sappony 230 kV	Active
AE2-040	Sapony 34.5 kV	Active
AE2-051	Carson-Septa 500 kV	Active
AE2-094	Carson-Rogers Road 500 kV	Active
AE2-122	Birdneck-Landstown 230 kV	Active

Queue Number	Project Name	Status
AE2-123	Birdneck-Landstown 230 kV	Active
AE2-124	Landstown 230 kV	Active
AE2-134	Locust Grove-Paytes 115 kV	Suspended
AE2-182	Briery-Clover 230 kV	Active
AE2-253	Hickory-Moyock 230 kV	Active
AE2-259	Curdsville-Willis Mtn 115 kV	Active
AE2-260	Clubhouse 230 kV	Active
AE2-313	Carson-Rawlings 500 kV	Active
AF1-069	Carson-Rogers Rd 500 kV	Active
AF1-079	Louisa-South Anna 230 kV	Active
AF1-123	Fentress 500 kV	Active
AF1-124	Fentress 500 kV	Active
AF1-125	Fentress 500 kV	Active
AF1-246	Clover-Rawlings 500 kV	Active
AF1-266	Clubhouse-Sapony 230 kV	Active
AF1-292	Fields 34.5kV	Active
AF1-301	Louisa-South Anna 230 kV	Active
AF2-037	Louisa-North Anna 230 kV	Active
AF2-042	Clover-Rawlings 500 kV	Active
AF2-063	Louisa-North Anna 230 kV	Active
AF2-081	Moyock 230 kV	Active
AF2-085	Midlothian 34.5 kV	Active
AF2-144	Powhatan 34.5 kV	Active
AF2-244	Powhatan 34.5 kV	Active
AF2-266	Clover 230 kV	Active
AF2-299	Fields 34.5 kV	Active
Z1-068	Birdneck 34.5kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P1-2: LN 594	CONTINGENCY 'DVP_P1-2: LN 594' OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 END
DVP_P1-2: LN 255-D	CONTINGENCY 'DVP_P1-2: LN 255-D' OPEN BRANCH FROM BUS 314232 TO BUS 957430 CKT 1 /* 8NO ANNA 230.00 - AF2- 037 TAP 230.00 END
DVP_P4-2: 2088T2135	CONTINGENCY 'DVP_P4-2: 2088T2135' /* GORDONVILLE 230 KV OPEN BRANCH FROM BUS 314745 TO BUS 314758 CKT 1 /* 6LOISACT 230.00 - 6GORDNVL 230.00 OPEN BRANCH FROM BUS 314734 TO BUS 314758 CKT 1 /* 6CASHSCORNER230.00 - 6GORDNVL 230.00 OPEN BRANCH FROM BUS 314734 TO BUS 314759 CKT 1 /* 6CASHSCORNER230.00 - 6HOLLYMD 230.00 OPEN BUS 314734 /* ISLAND: 6CASHSCORNER230.00 END
DVP_P1-2: LN 575	CONTINGENCY 'DVP_P1-2: LN 575' OPEN BRANCH FROM BUS 314911 TO BUS 314918 CKT 1 /* 8LADYSMITH 500.00 - 8NO ANNA 500.00 END
DVP_P4-2: H1T594	CONTINGENCY 'DVP_P4-2: H1T594' /* MORRISVILLE 500 KV OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 OPEN BRANCH FROM BUS 314063 TO BUS 314916 CKT 1 /* 6MORRSVL 230.00 - 8MORRSVL 500.00 OPEN BUS 314897 /* 8MORRS_1 500.00 KV END
DVP_P1-2: LN 2088	CONTINGENCY 'DVP_P1-2: LN 2088' OPEN BRANCH FROM BUS 314745 TO BUS 314758 CKT 1 /* 6LOISACT 230.00 - 6GORDNVL 230.00 END
DVP_P4-2: 568T575	CONTINGENCY 'DVP_P4-2: 568T575' /* LADYSMITH 500 KV OPEN BRANCH FROM BUS 314911 TO BUS 314922 CKT 1 /* 8LADYSMITH 500.00 - 8POSSUM 500.00 OPEN BRANCH FROM BUS 314911 TO BUS 314918 CKT 1 /* 8LADYSMITH 500.00 - 8NO ANNA 500.00 END

Contingency Name	Contingency Definition
DVP_P4-2: H1T575	CONTINGENCY 'DVP_P4-2: H1T575' /* LADYSMITH 500 KV OPEN BRANCH FROM BUS 314911 TO BUS 314918 CKT 1 /* 8LADYSMITH 500.00 - 8NO ANNA 500.00 OPEN BRANCH FROM BUS 314196 TO BUS 314911 CKT 1 /* 6LADYSMITH 230.00 - 8LADYSMITH 500.00 END
DVP_P4-2: SPOTS H1T594	CONTINGENCY 'DVP_P4-2: SPOTS H1T594' /* SPOTSYLVANIA 500 KV OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 OPEN BRANCH FROM BUS 314755 TO BUS 314934 CKT 1 /* 3SPOTSYL 115.00 - 8SPOTSYL 500.00 END

12 Light Load Analysis

Light Load Analysis is not required for solar projects.

13 Short Circuit Analysis

The following Breakers are overdutied:

None.

13.1 System Reinforcements - Short Circuit

None.

14 Stability and Reactive Power

To be determined in the Facilities Study Phase.

15 Affected Systems

15.1 TVA

None.

15.2 Duke Energy Progress

None.

16 Attachment 1: One Line Diagram

